Ref: 12009/3.17/1.0R/Bank A

Technical Manual

Installation Guide

Notice

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European Office
xxxxxxxxxxxxxxxxxxxxx

Tel: xxxxxxxxxxxxxxxxx

North American Office
xxxxxxxxxxxxxxxxxxxx,
xxxxxxxxxxxxxxxxxxxx

Tel: xxxxxxxxxxxxxxxx

Email: sales@companyx.com

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Revision History

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Preface

Document prepared for Bank A.

Operating System[1]

Unix

Database

Oracle

3rd Party System(s)

SunGard Global One

[1]Server operating system running `glassd`.

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Installation

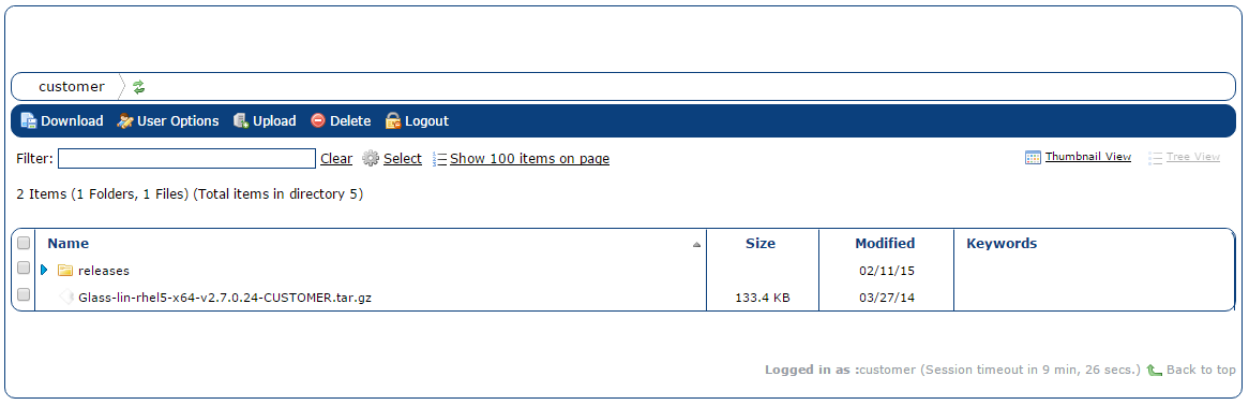
Software Download

The application is provided via electronic software delivery (ESD) from:

`https://ftp.companyx.com`

A username and password will be provided by Company X.

Once logged in you should see a web page[1] similar to:



The application software release archives are stored in the `releases` directory.

For example, Version 3.1.0.0 for Red Hat Enterprise Linux 6 64-bit x86 will be called:

`Glass-lin-rhel6-x64-v3.1.0.0.tar.gz`

The archive filename format is:

`Glass-<arch>-<version>.<format>`

where

- `arch` is the hardware architecture. For example, this will be `lin-rhel6-x64` for Red Hat Enterprise Linux 6 64-bit x86, or `win-x64` for 64-bit Windows.
- `version` is `Major.Minor.Revision.HotFix`. E.g. 3.1.0.0.
- `format` will be `tar.gz` for Unix or `.7z` for Windows.

Additional customer specific release archives will also be provided where necessary. The filenames of these archives contain a customer name suffix. These archives contain customer specific binaries, scripts and configuration files.

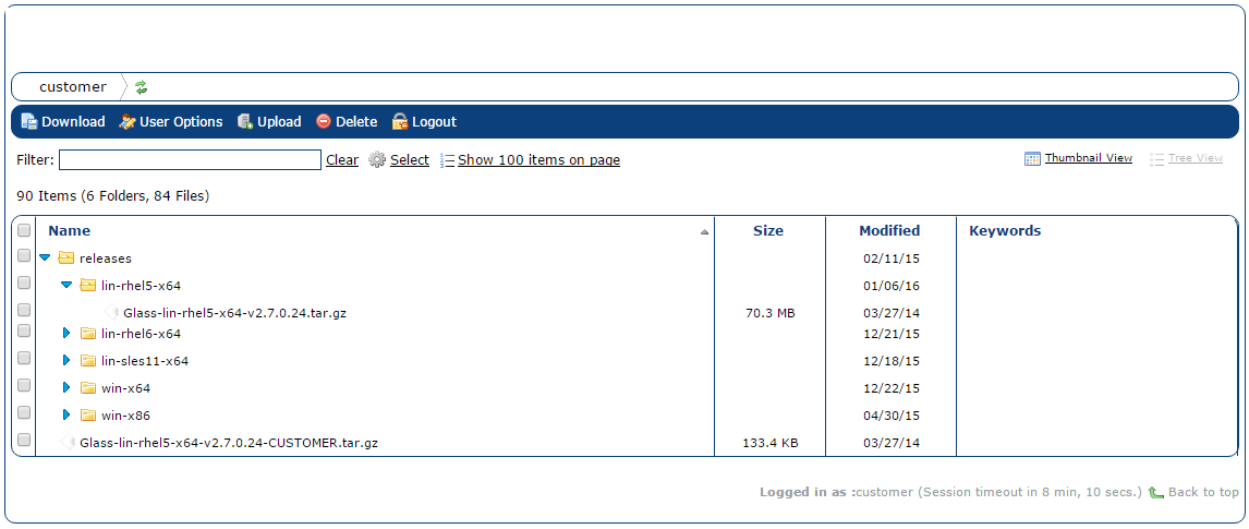
For example, a Version 3.1.0.2 release archive for Bank A (Unix) would be called:

`Glass-lin-rhel5-x64-v3.1.0.2-Bank A.tar.gz`

To commence the installation download and uncompress the distribution archives(s) for your chosen platform.

Within the `ftp.company.com` web interface check all files required and then click `Download`. For example:

[1]Other download methods are available. "Software Archive Download" on page 51 for details.



Release Archive Contents

Each primary release archive contains the following directories:

- `bin`. All binaries for the hardware architecture. Includes the executable `glassd` and all associated libraries.
- `configuration`. Common XML configuration files.
- `documentation`. Product documentation for end users and technical staff. Also includes licences for open source and other 3rd party components.
- `sql-scripts`. Database schema creation and migration scripts for the supported database products.
- `example-applications`. Example web services client source code for C#, Java and JavaScript (node).

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Hardware and Software Requirements

The hardware and software requirements are included in the Company X *Technical Overview* document in the Hardware Requirements and Software Requirements sections.

In addition to this, there are some specific requirements for Oracle Client Configuration. If Oracle and the application are not present on the same server, then the Oracle client software must be installed on the server where the application resides. Additionally, the version of the Oracle client must match the server version; for example, Oracle 11 client with an Oracle 11 server, Oracle 12 client with an Oracle 12 server, and so on. It is, however, possible to mix architectures so that a 32-bit Oracle client could connect to a 64-bit server (but not the other way around).

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Client Installation

No client side installation is required. The product is a web based application which runs in a web browser. It does not require any browser plug-ins to be installed although JavaScript must be enabled in the browser.

openfin

The application can be deployed and the client run within the openfin HTML5 container for financial applications.

This is achieved using the following steps:

1. Install the openfin runtime for Windows. Instructions on how to do this are provided on the openfin website:

   OpenFin Runtime Introduction

2. If you are an openfin customer an example `ta-openfin.json` file should be provided in your customer specific release archive (the release archive containing the suffix Bank A). If this file has not been provided then please contact `customer-service@companyx.com` for assistance.

3. Open the file `ta_openfin.json` in a text editor and substitute the protocol (<GLASS-PROTOCOL>), host (<GLASS-HOST>) and port (<GLASS-PORT>) placeholder variables for values which match your installation of the application and then save the file.

   E.g. change the line:

   ```
   "url": "<GLASS-PROTOCOL>://<GLASS-HOST>:<GLASS_PORT>",
   ```

   to

   ```
   "url": "http://localhost:10080",
   ```

   Note: if the example .json file contains comments of the form `//..` or `/*.. */` these should be removed as they are not allowed in JSON.

4. Launch openfin passing the location of the modified `ta-openfin.json` file in the `-config` parameter. E.g.

   ```
   openfin -launch -config openfin/ta-openfin.json
   ```


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Server Installation

The server consists of a binary executable process that uses XML configuration files and dynamically loaded modules. The process, `glassd`, contains a built-in HTTP server which users connect to using a web browser.

The server, when starting up, uses the configuration file, or files, to know which types of pages to deliver to the user. The configuration file contains references to modules which contain the functionality that is needed. These modules contain components such as the Borrower App, Position Exporter and the Net Account Position Browser screen.

An important part of the server architecture is that only the modules that are needed for a specific application need to be loaded. For example, if a deployment only interfaces to FIS/SunGard Martini/Apex and uses Microsoft SQL Server as a database the server does not need to load, or reference, the Global One, Oracle or MQ libraries. This avoids the need to install these 3rd party applications or libraries on the server.

To commence the installation copy the contents of both the primary release and customer specific archives to an installation directory of your choice.

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Initial Install

Within the customer specific release archive there will be a set of scripts to create an initial database. These scripts execute `glass-offline-loader` using the `glass-offline-loader.xml` configuration. This XML configuration file will specify a series of file loaders which reference the files in the `data-files` directory. To execute an initial install:

Change directory to the `client-examples/banka/initial-install` directory.

Copy initial data files into the `data-files` directory and run one of the `install-<database-name>.sh` files. For example, to create an initial Oracle database on Unix the script to run would be `install-oracle.sh`. To create an initial test SQLite database on Unix the script to run would be `install-sqlite.sh`.

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Starting the Server

To start `glassd` from the console run `glass.sh`.

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Permissions

The application executable, `glassd`, is a user space process and makes no use of any special kernel privileges. Because of this, the process should be run as a normal operating system user for security.

Also note that no operating system root access or administrator rights are required for installation. This is unless you want to integrate the server into your operating system start-up phase, `/etc/rc.local` for example.

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Licensing

Company X will supply a `companyx-licence.xml` file inside the customer specific release archive.

To view the current license information simply click on the main menu and then the About menu item. This will display the license information for each licensed application including the license expiry date. All open source component licenses are also included.

Applications which are not licensed can be used for a limited time. This will be noted in the information panel with a message similar to `'Evaluation Licence. Expiring in 2 hours.'`.

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Firewall Configuration

By default the `glassd` HTTP Server listens on port 10080. In production a web server typically runs on port 80 (HTTP) or 443 (HTTPS) although the server can be configured to listen on any port.

For details of how to modify the port used see HTTP Server.

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Scheduling and Shutdown

The `glassd` server process can be left to run for nearly 24 hours per day but must be shutdown and restarted at the beginning of each business day. This is to allow for the rolling forward of the current business date 'today'.

If the value of `environment/exit-at-midnight` is true then the server will automatically shutdown at 1 second to midnight (23:59:59).

If the value of `environment/start-of-day` is specified this will set the shutdown time to the specified value (in HH:MM format). This value, if specified, overrides `exit-at-midnight`.

The process should then be scheduled to restart at an appropriate time. It is suggested that this be after the majority of the nightly batch processes have completed but before users are expected to need to use the application.

If `glassd` is not configured to shut down itself, the process should be shut down by using the SIGTERM signal (allowing the application time to shut down gracefully), and only if it fails to do so send a SIGKILL.

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Overnight Process

The product has an overnight process requirement when interfacing to Global One. The provided configuration will include a directory called `overnight-process` which contains an XML configuration file for a `glass-offline-loader` which is run from `overnight-process.sh`.

The XML configuration file references a loader configuration for 'G1Trade' and specifies the fields in the Global One Position File Control Code. This same control code is also used by the intra-day trade stream. The reason why the same control code is used is to ensure that Global One clears the control code when the file is processed and when the stream starts up the 'pipe' is empty.

The typical schedule is as follows:

- `glassd` shuts down at the end of day (for example, 5:30pm EST).
- Global One day roll occurs.
- An extract using the Position Control Code runs to generate the daily reconciliation file.
- `overnight-process.sh` is run to load the extract into the application.
- `glassd` is restarted and connects to the control code in Global One (via GCI) and listens for trade updates.

For more information about the Global One Position File see the Company X *G1 Integration Installation Overview* document and the Global One GOSOS manual.

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Configuration

Under normal circumstances Company X will supply an installation configuration, which will require minimal modification prior to deployment, within the customer specific release archive downloaded from the Company X website.

The purpose of this section is to document the parts of the configuration which are most likely to be modified by a customer; a detailed explanation of every aspect of configuration is beyond the scope of this document. If you require assistance with aspects of server configuration not covered by this document please contact Company X Support.

Editing Configuration Files

All of the configuration files are XML files. It is therefore advisable to edit the files using an editor which provides syntax highlighting of XML files. This will make viewing and editing the files much easier. Suggested editors include Notepad++ (Windows), TextPad (Windows) and Sublime Text (Windows, Linux).

Changes to the configuration file(s) are only recognised when `glassd` is started or restarted.

glass.xml

`glass.xml` is the main configuration file which is loaded first by the `glassd` server. More often than not this XML file references, or includes, other XML files.

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Includes

The configuration XML files support a file inclusion mechanism. Within the `glass.xml` or any other XML file referenced from `glass.xml` it is possible to specify that another XML file should be included in-line within the current file. For example:

```
<include filename = 'system/environment.xml' />
```

Typically a configuration will be split into separate XML files where each file deals with a specific aspect of the server configuration such as Authentication or SMTP.

<environment> Element

An important part of the `glass.xml` configuration is the `environment` element.

For example, to configure the HTTP server listening port the environment element may contain:

```
<httpd>
        <port value='10081' />
</httpd>
```

or (the equivalent):

```
<httpd>
        <port>10081</port>
</httpd>
```

This value can, however, be overridden on the command line using:

```
-httpd/port=11000
```

The use of the forward slash means that this refers to the `port` element within the `http` element.

Environment Variables

When the configuration is loaded all of the name value pairs within the `environment` element are added to system environment variables along with names and values specified on the command line.

For example, this element within the `environment` element defines an environment variable:

```
<smtp-server-port value='25' />
```

It is equivalent to a system environment variable called `SMTP-SERVER-PORT` or a command line parameter of:

```
-smtp-server-port=25
```

Environment Variable Substitution

Another feature of the server configuration is that environment variables can be used within the XML files. When the configuration files are read the environment variable's value will be substituted for `$(VARIABLE_NAME)`.

For example, if the environment variable `SMTP_USERNAME` is set to `ADMIN` then:

```
<smtp-server-username value='$(SMTP_USERNAME)' />
```

will become:

```
<smtp-server-username value='ADMIN' />
```


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Note that if the value of an environment variable is specified on the command line it will override a system environment variable's value with the same name. Likewise a command line parameter will override both a hardcoded value within the `environment` element or system environment variable value for the same name.

The order of precedence is:

1. Command line parameter
2. Environment XML element
3. System environment variable

The environment variable substitution is case insensitive so `$(SMTP_USERNAME)` is the same as `$(smtp-username)`.

If an environment variable hasn't been given a value then a warning message will be written to the log output. For example:

```
20140320 11:56:25+0000 + [  1] Environment variable not found: 'SMTP_USERNAME'
20140320 11:56:25+0000 + [  1] Key evaluated to: 'smtp_username'
```

Environment variable $ references can also include forward slashes. For example, the value `mart_p1` in:

```
<environment>
       <martini>
               <database>mart_p1</database>
```

could be referred to using `$(martini/database)`.

Conditional Includes

When environment variable substitution is used in conjunction with file includes it is possible to conditionally include other configuration files. For example:

```
<include filename = 'environments/$(GLASS_ENVIRONMENT)/environment.xml' />
```

will include a different configuration file depending upon the value of the environment variable `GLASS_ENVIRONMENT`.

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Command Line Parameters

The server supports various command line parameters which cannot be specified elsewhere such as in the `glass.xml` configuration file. The command line to start `glassd` might look something like this:

```
glassd -configuration-file=configuration\glass.xml -logging-level=2 -log-filenamebase=logfiles\glass
```

Command line parameters must be prefixed with a minus sign ('-').

Parameter	Description
configuration-file	By default the configuration file `glass.xml` will be loaded from the current working directory. This parameter can contain the relative path of the configuration file within the current working directory.
today	By default the application runs with the value of 'today' set to the current system date. This can be overridden using this parameter. Format CCYY-MM-DD or -1 (previous business day).
start-of-day	Specifies the start of day time. Format HH:MM. Sets the shutdown time to the specified time. If the time is prior to the current time then the shutdown is set to the next day at the specified time. For example, with a `start-of-day` of 18:00 starting the server at 18:05 on 13th March would advance the internal date by 1 day and set the shutdown to 18:00 on 14th March.
log-filesize	The maximum size of a log file in MB. If not specified defaults to 0 (no maximum size). Specifying this command line parameter will append a number to the log filename starting with "00000000". A new log file will be created with this number incrementing by one when the maximum log file size is reached. Note that the number resets to "00000000" each time `glassd` is executed.
log-filename	Specifies the format of log file names. Defaults to '_YYCCMMDD_HHMMSS' if `-logfilenamebase` is specified.
log-filenamebase	Specifies the prefix of the log file names when `-log-filename` is not specified. Defaults to an empty string.
working-directory	If specified the current working directory is changed to the given directory, if possible. Typically used when the application is started as a Windows service and not from a batch file.
install-service	Installs the application in the Service Control Manager (SCM). Windows only.
uninstall-service	Uninstalls the application from the Service Control Manager (SCM). Windows only.
service	Specified when installing the application as a service. Also ensures that log output is also written to the Windows Event Log. Windows only.
service-name	The name of the service in Service Control Manager (SCM). Required when installing (-install-service) and uninstalling a service (-uninstall-service). Windows only.
validate-installation	If this option is specified, then a check is performed to verify that all resource identifiers assigned to roles are registered. If the validation fails, then this is shown in the log, along with the message, `Installation failed validation due to the errors shown above`.

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Authentication

The application can be configured to authenticate users from its own internal database or integrate with Third Party applications and services such as FIS/SunGard Martini/Apex or Windows Active Directory. The authentication configuration element is used to configure how the application implements authentication, whether Forms Based Authentication or HTTP authentication is used and what systems (Login Providers) the authentication layer interfaces with. The authentication element is structured as follows:

```
<authentication access-model     = '$(db-access-model)'
                database         = 'glass-db'>
    <allowed-mechanisms>
        <fba />

        <http />
    </allowed-mechanisms>

    <providers>

    </providers>
</authentication>
```

The `authentication` element is comprised of two sub-elements, `allowed-mechanisms` and `providers`. The `allowed-mechanisms` element is used to configure Forms Based Authentication and HTTP Based Authentication via the `fba` and `http` elements respectively. The `providers` sub-element is used to define configuration settings for each Login Provider. Please note that not all Login Providers require specific configuration.

The `fba` element is used to define whether Forms Based Authentication should be used by the application and which Login Providers are used to implement it. For example, the following example shows how the application would be configured to use FIS/SunGard Martini/Apex and the Internal Database for Forms Based Authentication:

```
        <fba        providers       = 'martini,internal' />
```

The `http` element is used to specify whether HTTP Based Authentication should be used by the application, which HTTP Authentication Mechanisms should be used and which Login Providers are used to provide the services. For example:

```
        <http>
            <Basic      providers = 'internal' >
            <Digest     providers = 'windows' >
            <NTLM       providers = 'windows' >
            <Negotiate  providers = 'windows' >
        </http>
```

Not all providers support HTTP Authentication and not all providers that support HTTP authentication support all mechanisms.

In order to support Single Sign On the NTLM and/or Negotiate mechanisms should be used. If Basic and Digest mechanisms are used, then the user will be prompted for their login credentials by the browser.

Login Providers

The providers' attributes in both the `fba` and `http` elements can be used to specify one or more Login Providers. The currently supported list of Login Providers for your platform is as follows:

- Internal authentication.
- PAM. Unix based Pluggable Authentication Module.
- Kerberos. Kerberos Authentication.

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As already mentioned the `providers` sub-element can be used to implement provider specific configuration settings which are detailed below.

Internal

There are no specific settings for the Internal Login Provider.

LDAP

LDAP (Lightweight Directory Access Protocol) is an Internet protocol that applications can use to lookup information about users and groups. An LDAP server, such as a Windows Active Directory Server or an OpenLDAP Server, contains a collection of data about users and groups that can be used by an application that supports the LDAP protocol to, for example, perform user authentication. A typical LDAP configuration element would look something like the following:

For authentication against an OpenLDAP server:

```
<ldap>
<!-- For OpenLDAP -->
<server address = 'openldap.example.com'
port = '389'
domains = 'EXAMPLE'
auth-mechanism = 'SIMPLE'
user-bind-dn-format = 'uid=%u,ou=Users,dc=example,dc=com'
use-protocol-version = '3'
enable-openldap-debug-logging = 'true'
/>
</ldap>
```

Or for authentication against a Microsoft Active Directory server :

```
<ldap>
<!-- For MS AD -->
<server address = 'activedirectory.example.com'
<!-- port 389 or 3268 depending on AD configuration -->
port = '3268'
domains = 'EXAMPLE'
auth-mechanism = 'SIMPLE'
user-bind-dn-format = '%d\%u'
use-protocol-version = '3'
/>
</ldap>
```

The meaning of these attributes is as follows:

`address`. This is the address of the LDAP server to connect to for authorisation requests. For example, ldap.example.com.

`port`. The port number the directory server is listening on. If this is left blank the default port of 389 will be used.

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`domains`. A comma separated list of domain names. This list of domains will be displayed in the application login dialog inside the 'Domain' drop-down. Example domains would include `EXAMPLE.COM` and `COMPANY.LOCAL`.

`auth-mechanism`. This should be set to `SIMPLE` for OpenLDAP and for Windows Active Directory set to SIMPLE or GSSAPI.

`user-bind-dn-format`. Optional and default is blank. It is a format string for the UserDN. Replaceable placeholders are %u for user, %d for domain, %r for domain/realm.

`use-protocol-version`. Optional and default is 3.

`enable-openldap-debug-logging`. Optional and default is false.

PAM

PAM (Pluggable Authentication Modules) is a common Unix mechanism that allows applications to authenticate via PAM. PAM can be configured to authenticate via local or remote identity management services.

Through PAM, the application can authenticate its users with the local Unix user accounts (i.e. as listed in `/etc/passwd`).

A typical PAM configuration for Unix would look something like :

```
<pam service-name = 'su' />
```

The `service-name` attribute is used to specify the PAM service as defined in `/etc/pam.d/` (Linux), or `/etc/pam.conf` (Solaris).

Under Linux, the default PAM service "su" should be sufficient for authenticating as a local unix user account.

Under Solaris, the default PAM service "" (will default to using the service "other") and this should be sufficient for authenticating as a local unix user account.

Note : For PAM to authenticate local Unix user accounts, the process invoking PAM needs read access to the /etc/shadow file. Therefore the glass process will need read access to this file. This can be achieved via creating a user and group for "glass", and assigning read permission to the "glass" group e.g.

```
$ useradd glass
$ chown root:glass /etc/shadow
$ chmod 440 /etc/shadow
```

A listing of the file should then look like:

```
$ ls -la /etc/shadow
-r--r----- 1 root glass 1834 Jul 24 14:22 /etc/shadow
```

Your System Administrator should be consulted to implement read permission on `/etc/shadow`.

PAM is a configurable mechanism and can also be used to authenticate to other services such as LDAP (e.g. Windows Active Directory), Kerberos, SSS.

Kerberos

The Kerberos Login Provider can be used to integrate with either a Windows Active Directory server or a Unix based Kerberos server.

A typical Kerberos configuration element for Unix would look something like:

```
<kerberos    config-file   = 'configuration/environments/common/krb5.conf'
             keytab-file   = 'configuration/environments/common/krb5.keytab'
             service-name   = 'HTTP/server001.example.com@EXAMPLE.COM'
/>
```

The meaning of these attributes are as follows:

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`config-file`. This is the Kerberos krb5.conf file which is typically located within the file heirarchy rather than the system `/etc/krb5.conf` file.

`keytab-file`. This is the Kerberos keytab file for this service as specified in the `service-name` attribute.

`service-name`. This is the Kerberos service name for this process as used in the generation of the keytab file. Typically this includes the protocol, the FQDN of the server and the NetBIOS name.

To generate a Kerberos keytab file to match the following example configuration:

For a Microsoft Active Directory server:

```
C:\> ktpass -princ HTTP/server001.example.com@EXAMPLE.COM -mapuser IT@EXAMPLE.COM -pass password -
crypto RC4-HMAC-NT -ptype KRB5_NT_PRINCIPAL -out C:\server001.keytab
```

Please note there is a known bug with some versions of ktpass.exe where a –pass * parameter is used. This parameter means the user will be prompted for a password, however it can result in the user password being changed unexpectedly. See knowledge base articles:

http://support.microsoft.com/en-us/kb/kbview/939980

http://support.microsoft.com/en-us/kb/960830

Before the –pass * parameter is used with ktpass.exe, you should ensure you apply the appropriate hotfix version to Windows Server which resolves the bug.

For a Unix Kerberos server:

```
$ kadmin.local
> ktadd -k server001.keytab HTTP/server001.example.com@EXAMPLE.COM
```

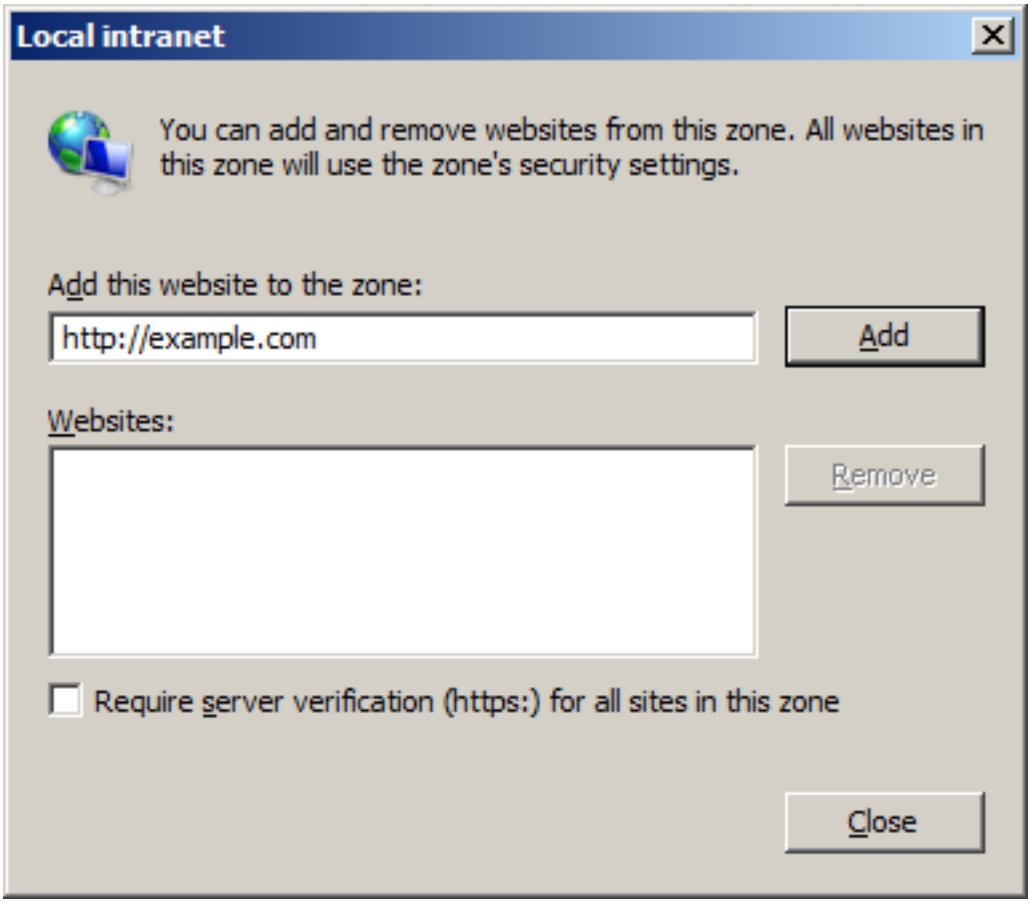
Then copy the output file `server001.keytab` to the directory `configuration/environments/common/krb5.keytab`

Browser Configuration

If Single Sign On is enabled, the user's web browser may need to be configured to allow Integrated Authentication. The method varies between browsers and may vary slightly depending on the browser version.

Internet Explorer

- Go to Settings -> Internet Options
- From the Internet Options dialog select the Security tab
- Click on the Local intranet icon to enable the Sites button
- Click the Sites button
- Once the Local internet dialog opens, press the Advanced button
- Add your site e.g. http://example.com to the list of Local intranet websites



Chrome

Integrated Authentication is only enabled when Chrome receives an authentication challenge from a proxy or when it receives a challenge from a server which is in the permitted list.

Chrome can inherit a list of permitted servers from the Local Intranet websites configured under IE as described above. Alternatively a list of websites can be passed to chrome using a command-line switch.

The list is passed to Chrome using a comma-separated list of URLs via the `auth-server-whitelist` command-line switch. For example, you can pass in: `--auth-server-whitelist="*example.com,*example.org"` which would tell Chrome that any URL ending in either 'example.com' or 'example.org' is in the permitted list. Without the '*' prefix, the URL has to match exactly.

If the command-line switch is not present, then in Windows only the permitted list consists of those servers in the Local Machine or Local Intranet security zone (for example, when the host in the URL includes a "." character it is outside the Local Intranet security zone), the same as in IE.

Firefox

Firefox may prompt for Windows Username and Password rather than obtaining it from your Windows Credentials. Single Sign On can be enabled if the `about:config network.negotiate-auth.trusted-uris` setting is changed to include the url of the server:

- Open Firefox and type `about:config` in the address bar
- Scroll down to the `network.negotiate-auth.trusted-uris` setting

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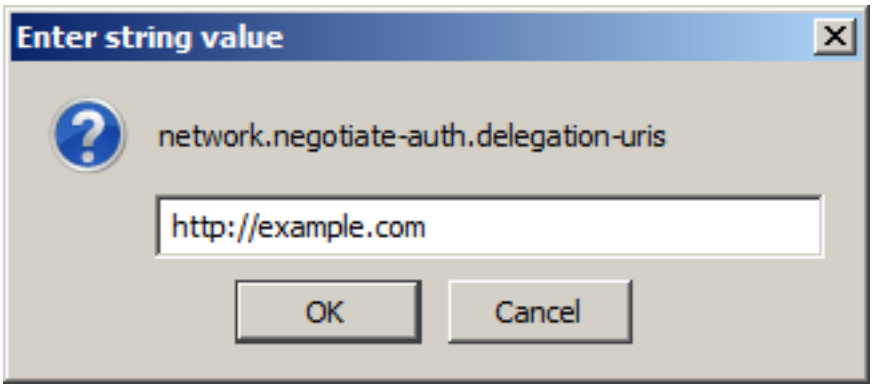
- Right-click on `network.negotiate-auth.trusted-uris` and choose modify
- Enter the url of your site e.g. http://example.com

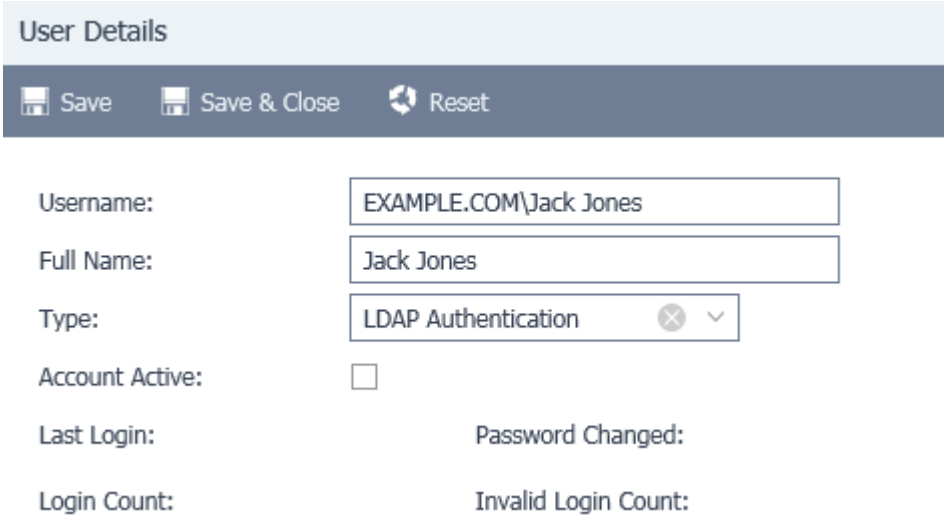


User Setup

Within the application individual users will need to be set up and configured to indicate which authentication mechanism they are using.

If the user account is a Windows Active Directory based account then the domain name must be configured as part of the username in the format `DOMAIN\user`. When the server that the application is running on is Unix based this should be the full NetBIOS domain name. For example `EXAMPLE.COM\user`.

When creating a user specify the Authentication Type as 'Glass Authentication', 'Unix (PAM) Authentication' or 'Kerberos Authentication':



A password is not required in this dialog when PAM or Kerberos Login Providers are used as the password authentication is handled by the back-end server.

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HTTP Server

Thread Pooling

It is possible to tune the built-in HTTP server thread pool. The thread pool is used to service client requests.

By default the thread pool contains 10 threads (for both 64-bit Windows and Linux). In the current version of the application the thread pool size does not automatically increase in size so, in some circumstances, it may be necessary to manually increase the pool size.

To modify the thread pool size set a value for `environment/httpd/threadcount`.

Port and NIC

To modify the listening port used specify a value for `environment/httpd/port`. If this isn't specified the default port will be 10080.

By default the server will listen on any local NIC (Network Interface Card). To use a specific NIC set `environment/httpd/nic` to its IP address.

For example, to listen on 192.168.0.1:80:

```
<environment>
        <httpd>
                <port>80</port>
                <nic>192.168.0.1</nic>
        </httpd>
```

It is common practice for an HTTP reverse proxy server to sit in front of glassd and direct traffic from the user-facing port 80 to glassd's port 10080 (or alternative). See section HTTP Proxy Server for more details.

Session Timeout

By default the built-in HTTP Server will timeout each HTTP session after 15 minutes of inactivity. This timeout can be modified by specifying a `session-timeout` element within the `environment` element of `glass.xml`. The value of the element is the session timeout in minutes. For example, to set the session timeout to 3 hours (180 minutes) specify:

```
<environment>
        <httpd>
                <session-timeout value='180' />
        </httpd>
```

CSRF Checks

By default, CRSF (Cross-Site Request Forgery) checks are enabled. It is possible to disable CRSF checks to facilitate testing. This is achieved by specifying a `csrf-bypass-check` element within the `environment` element of `glass.xml`:

```
<environment>
      <httpd>
              <csrf-bypass-check value='true' />
      </httpd>
```

In which case, there will be no failures from the login attempt due to CSRF checks, and the server will log "CSRF bypass enabled in HTTPSessionManager. Suspected cross site request forgery attempts will not be logged."

Note that this bypass should only be used in a test (usually QA or acceptance test) environment, and never in Production.

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Login Restriction

By default a user can login with their account from multiple different browsers simultaneously. Some organisations wish to restrict logins to a single concurrent session per user. This is achieved using:

```
<environment>
	<httpd>
		<multiple-logins value='false' />
	</httpd>
```

SSL / HTTPS

The `glassd` built-in web server supports the HTTPS protocol for SSL encrypted communication between the user's web browser and the server. To use HTTPS an SSL certificate must be obtained and configured in the configuration file.

The application uses a single certificate file for the server certificate, any intermediary CA certificates (if applicable) and the root CA certificate. The file must be in PEM format with the certificates listed in the following order:

- Server Certificate
- Intermediate CA Certificates
- Root CA Certificate

The certificate can be a self-signed certificate or a Certificate Authority generated certificate, however if a self-signed certificate is used then users may be warned that there are potential security risks unless the self-signed certificate is installed as a trusted certificate on every client machine that accesses the application. Consequently, whilst self-signed certificates are ideal for test environments they are not recommended for production.

For details of how to generate self signed or Certificate Authority certificates see Certificate Generation.

Both the certificate and private key files must be saved to the `configuration/environments/common/` directory. The file names and passphrase should then be included in a `ssl` element within the `environment/httpd` element of `glass.xml`. For example:

```
<environment>
	<httpd>
		<ssl>
			<version value='6' />
			<cipher-list value='ALL' />
			<certificate value='configuration/environments/common/server001.example.com.crt' />
			<private-key value='configuration/environments/common/server001.example.com.key' />
			<passphrase value='example-passphrase' />
		</ssl>
	</httpd>
</environment>
```

The 'version' must be a number from the table below :

SSLVERSION_DEFAULT	0
SSLVERSION_NEGOTIATE_HIGHEST	1
SSLVERSION_SSLv2	2
SSLVERSION_SSLv3	3
SSLVERSION_TLSv1_0	4
SSLVERSION_TLSv1_1	5
SSLVERSION_TLSv1_2	6

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The 'cipher-list' comes from the list pertinent to the selected protocol (as documented at https://www.openssl.org/docs/manmaster/apps/ciphers.html), where each cipher suite is colon separated e.g. 'ALL:ADH:EXP:MD5:@STRENGTH'.

The version, cipher-list, and passphrase configuration items are optional; certificate and private-key are **mandatory**.

Using HTTPS does not change the default port the HTTP Server listens on. For details of how to modify the port used see HTTP Server.

It is common practice for an HTTP reverse proxy server to sit in front of glassd and direct traffic from the user-facing port 80 to glassd's port 10080 (or alternative). In this situation it is best to let the reverse-proxy apply the HTTPS. See section HTTP Proxy Server for more details.

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HTTP Proxy Server

The `glassd` process contains a built-in web server to provide the user interface. By default this listens on port 10080 but can be configured to listen on typical web ports such as 80 or 443. As only one application can listen on a port, if the hosting machine contains other web-oriented applications then a HTTP reverse-proxy server may be used to avoid multiple usage of the ports 80/443. The Proxy Server can listen on port 80 and then route traffic to each of the applications on their unique ports e.g. glass default listening port is 10080. In addition, the Proxy Server can apply HTTPS/SSL. A typical reverse Proxy Server is Apache. An example configuration for this can be made in its httpd.conf (or equivalent file) for the machine host.example.com:

```
# Enable these modules
LoadModule proxy_module modules/mod_proxy.so
LoadModule proxy_http_module modules/mod_proxy_http.so

# For v3 and above, we need to proxy the websocket traffic
LoadModule proxy_wstunnel_module modules/mod_proxy_wstunnel.so

# We avoid use of forward proxying with "ProxyRequests off"
ProxyRequests off
# Ensure the network timeout is a sensible amount for long-running requests
ProxyTimeout 1200

# Proxy all traffic with root path "/ta" to glassd listening on port 10080
ProxyPass /ta http://localhost:10080/ta retry=30
# Also proxy websocket traffic in v3 and above
ProxyPass /ta  ws://localhost:10080/ta retry=30
ProxyPassReverse / http://localhost:10080/
ProxyPassReverseCookieDomain localhost host.example.com
```

This configuration can also be easily combined with the Apache virtual hosts feature, and also HTTPS/SSL as in the example below:

```
<VirtualHost *:443>
       ServerName host-virt01.example.com:443
       ProxyRequests off
       ProxyTimeout 1200
       # Proxy all traffic with root path "/ta" to glassd listening on port 10080
       ProxyPass /ta http://localhost:10080/ta retry=30
       ProxyPass /ta  ws://localhost:100801/ta retry=30
       ProxyPassReverse / http://localhost:10080/
       ProxyPassReverseCookieDomain localhost host-virt01.example.com

       #   Enable SSL for this virtual host.
       SSLEngine on
       # Disable undesired protocols e.g. SSLv2
       SSLProtocol all -SSLv2
       #   SSL Cipher Suite:
       # List the ciphers that the client is permitted to negotiate.
       # See the mod_ssl documentation for a complete list.
       SSLCipherSuite ALL:!ADH:!EXPORT:!SSLv2:RC4+RSA:+HIGH:+MEDIUM:+LOW

       # May need to add in a CA certificate ...
       # SSLCaCertificateFile /etc/httpd/conf/gs_intermediate_ca.crt

       SSLCertificateFile    /etc/httpd/conf/host-virt01.example.com.crt
       SSLCertificateKeyFile /etc/httpd/conf/host-virt01.example.com.key

       # Fix the cookie path and secure it
       Header edit Set-Cookie "^(.*)" $1;Secure
</VirtualHost
```


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E-mail (SMTP)

The `glassd` server has the ability to send emails, both automatically and manually triggered by a user.

E-mails may be sent automatically following the success or failure of an automated task run (such as a data file import or export). Manually triggered emails are sent when users request a data export or request a password reset.

There are two parts to `glassd` SMTP configuration 1) backup e-mailer 2) postbox.

Backup E-mailer

The backup e-mailer is used by the `glassd` task manager if the e-mail postbox configuration (below) should fail or when a serious connectivity error occurs.

To configure the backup e-mailer modify the SMTP settings within the `environment/smtp` element:

```
<smtp>
       <server value='smtp.example.com' />
       <server-port value='587' />
       <server-username value='$(smtp-server-username)' />
       <server-password value='$(smtp-server-password)' />
       <from value='no-reply@example.com' />
       <to value='' />
</smtp>
```

The meaning of the elements is as follows:

`server`. The hostname of the SMTP server or gateway.

`server-port`. Port 25 is usually used where authentication is not required and the server is accessible over the Internet. Port 587 is often the preferred port for authenticated e-mail submission within an organisation. Port 465 would be specified for SMTPS (SMTP with SSL).

`server-username` and `server-password`. If the SMTP server accepts anonymous connections then `server-username` and `server-password` could be left blank. If user credentials are required then it is recommended that a system e-mail account is used.

`from` should contain the from e-mail address. Typically, this will be a 'no-reply' e-mail address as the e-mail account will likely not be monitored.

`to` is typically left blank as the destination e-mail address is likely specified elsewhere.

Advanced Options

`use-tls`. If Transport Layer Security (TLS) is required specify 'true'. By default this is 'false' and TLS is not used. If TLS is used the `server-port` is usually 25 or 587. The additional elements `<ca-path>` or `<ca-file>` are used with `<use-tls>` to specify certificate details.

`disable-html`. If specified this regular expression is used to filter out HTML content in any e-mails which have a 'to' e-mail address matching the regular expression.

Postbox

The primary e-mail configuration is setup within the application itself rather than the XML configuration file.

Log in with an admin account and open the Task Management app.

Right-click the 'Outbound EMail Sender' row, and select *Edit…* to display the *Details* tab:

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Type:	EMailSend
Identifier:	OutboundEMail
Title:	Outbound EMail Sender
Description:	Test
Owner:	System Administrator
Logging Type:	Failures Only
Success Notify Destination:	Select a pre-defined destination
Failure Notify Destination:	Select a pre-defined destination

To add or edit a postbox, select *Postbox* to display the *Postbox* tab. From the *Postbox* dropdown, select the appropriate postbox (e.g., SMTP Gateway). To edit the postbox settings, click the pencil icon in the Postbox dropdown to display the *E-Mail Postbox Editor*. These postbox settings are likely to be similar to those in the XML configuration file `smtp` element:

E-Mail Postbox Editor

Description:	SMTP Gateway
Transport:	SMTP
Server:	smtp.tradingapps.com
Port:	587
EMail:	no-reply@example.com
Username:	username
Password:	••••••••••••••••
Use SSL:	☐
Domain:	
Maximum Retry Count:	10
Message Rate Limit:	120

CANCEL SAVE

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The 'Maximum Retry Count' is the maximum number of times the Outbound EMail Sender will attempt to send an e-mail before reporting a failure.

Some SMTP servers, including Microsoft Exchange, limit the rate at which e-mail messages can be sent. Exchange may generate the following SMTP error message if the rate is too high:

```
421 4.4.2 Message submission rate for this client has exceeded the configured limit
```

The 'Message Rate Limit' specifies the maximum number of e-mail messages which will be sent whenever the Outbound EMail Sender task executes. This can be left blank (no limit) or set to a value in order to avoid the 421 4.4.2 error.

E-mail (POP3)

The `glassd` server has the ability to receive emails.

There is only one part to `glassd` POP3 configuration, a mailbox.

```
<pop3>
       <server value='pop3.example.com' />
       <server-port value='110' />
       <server-username value='$(pop3-server-username)' />
       <server-password value='$(pop3-server-password)' />
</pop3>
```

The meaning of the elements is as follows:

`server`. The hostname of the POP3 server or gateway.

`server-port`. Port 110 is usually used where authentication is not required and the server is accessible over the Internet. Port 995 would be specified for POP3S (POP3 with SSL).

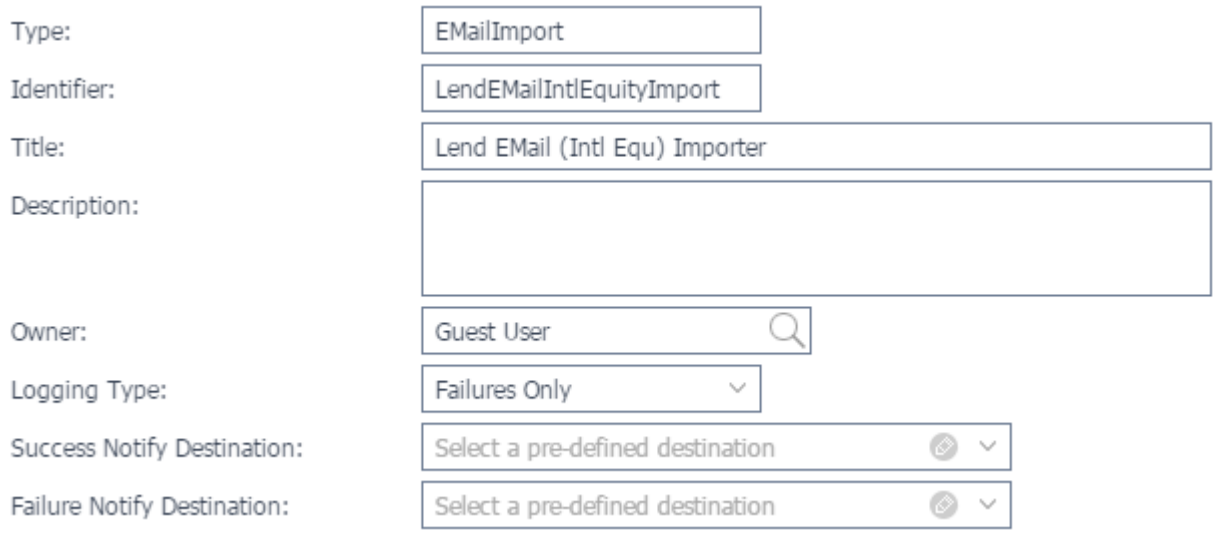
`server-username` and `server-password`. If the POP3 server makes anonymous connections then `server-username` and `server-password` could be left blank. If user credentials are required then it is recommended that a system e-mail account is used.

Mailbox

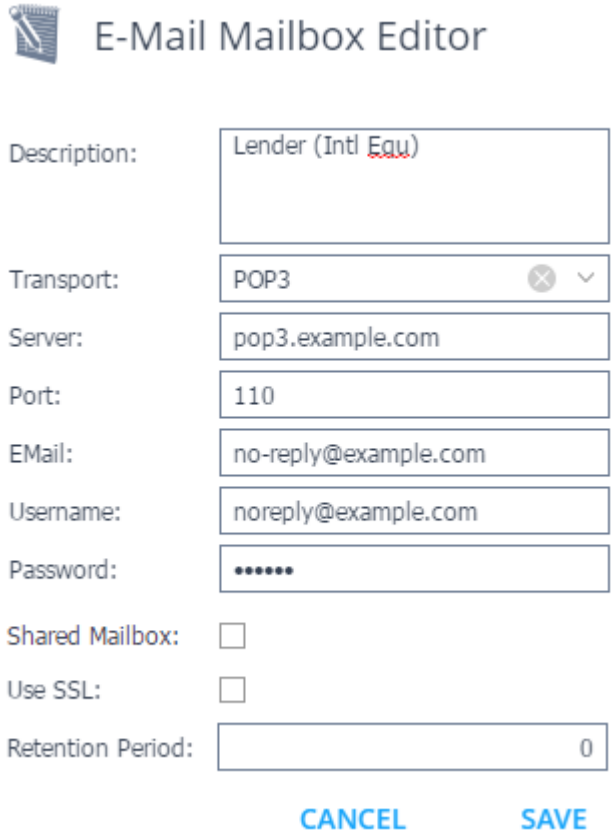
The primary e-mail configuration is setup within the application itself rather than the XML configuration file.

Login with an admin account and open the Task Management app.

Right-click, for example, the 'Lend Email (Intl Equ) Importer' row, and select *Edit...* to display the *Details* tab:



To add or edit a mailbox, select *Mailbox* to display the *Mailbox* tab. From the *Mailbox* dropdown, select the appropriate mailbox. To edit settings, click the pencil icon in the *Mailbox* dropdown to display the *E-Mail Mailbox Editor*. These mailbox settings are likely to be similar to those in the XML configuration file `pop3` element.



Using Shared Mailboxes

If you want to specify a shared mailbox / folder and a user name, and your POP3 server supports 'SASL PLAIN' then note the following:

`Username` is the service account (e.g., companyx/TestDelegate or TestDelegate@companyx).

`Email` is the email address of the shared mailbox / folder, (e.g., TestShared@companyx.com).

`Shared Mailbox` should be selected.

Make sure that the email addresses of the shared mailboxes are present in the configuration under lend-email-listeners, for example:

```
<lend-email-listeners value='TestShared@companyx.com,TestShared_
2@companyx.com,TestShared_3@companyx.com' />
```

The importer will then log in to POP3 as companyx/TestDelegate, specifying the mailbox TestShared, and expecting emails sent to TestShared@companyx.com.

E-mail SSL Configuration

Both SMTP and POP3 support SSL, enabled by checking the "Use SSL" checkbox in the appropriate Editor.

The configuration items are located in the `glass.xml` file :

SMTP

```
<smtp>
       <ca-path                value='<path-to-file>' />
       <ca-file                value='<fully-qualified-filename>' />
       <ca-allow-self-sign     value='false' />
</smtp>
```

POP3

```
<pop3>
       <ca-path                value='<path-to-file>' />
       <ca-file                value='<fully-qualified-filename>' />
       <ca-allow-self-sign     value='false' />
       <verbose                value='true' />
       <ssl-version            value='6' />
       <ssl-cipher-list        value='SHA1' />
       <ssl-cert-type          value='PEM' />
</pop3>
```

Where:

`ca-path` is the full path to the `ca-file` (see below), for example, `/companyx/release/companyx-server/latest/configuration/secure`. Remove if `ca-allow-self-sign` (see below) is set to `true`.

`ca-file` is the fully qualified, absolute file name of the Certificate Authority certificate, for example, `/companyx/release/companyx-server/latest/configuration/secure/exchange-test-pop3.pem`. Remove if `ca-allow-self-sign` (see below) is set to `true`.

`ca-allow-self-sign` species whether or not to allow self-signed certificates. In this case, `false` prevents use of such certificates. If set to `true`, remove `ca-path` and `ca-file` above. If a self-signed certificate is used then there are potential security risks unless the certificate is installed as a trusted certificate on every client machine that accesses the application. Consequently, whilst self-signed certificates are ideal for test environments they are not recommended for production.

`verbose` specifies whether or not to display extra details about the operations being performed. If the logging is configured with a value of `true`, then this will display verbose logging information for all POP3 communication inside `glassd`. When using SSL, the `server` value in `glass.xml` must match the server configured in the task, and hence also the server name in the certificate.

`ssl-version` here is `6`, representing SSLVERSION_TLSv1_2. Other possible values are:

0= SSLVERSION_DEFAULT

1= SSLVERSION_NEGOTIATE_HIGHEST

2= SSLVERSION_SSLv2

3= SSLVERSION_SSLv3

4= SSLVERSION_TLSv1_0

5= SSLVERSION_TLSv1_1

`ssl-cipher-list` is a list of one or more cipher strings separated by colons. The actual cipher string can take several forms, coming from the list pertinent to the selected protocol (for more see https://www.openssl.org/docs/manmaster/man1/ciphers.html). In the configuration above, `SHA1` represents all cipher suites using the digest algorithm SHA1.

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`ssl-cert-type` is the SSL certificate type, in this case, `PEM` (Privacy Enhanced Mail) is used for different types of X.509v3 files that contain ASCII data. This is the standard format for OpenSSL and many other SSL tools. This format is designed to be safe for inclusion in ASCII or even rich-text documents, such as emails.

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Global One Client Interface (GCI)

EXTRT/ACTIV Message Types

By default the GCI ACTIV message type is used to retrieve activity and position details. It is possible to override this behaviour and use the EXTRT message type instead if the Global One version supports EXTRT/EXTRA message types (support for EXTRT/EXTRA message types via GCI Sockets was added in Global One 7.3.0). This can be achieved using an environment variable:

```
<gci-extrt-supported value='true' />
```

or the equivalent command line parameter:

```
-gci-extrt-supported=true
```

The EXTRT/EXTRA and ACTIV/ACTIA message types are functionally equivalent in Global One.

For further information about GCI and the GCI Message Types see the SunGard Global One Client Interface (GCI) manual.

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Data Importers

Within the supplied configuration files there will likely be a series of loader (file import) definitions. An example loader definition would look like:

```
<FINSecurityImport
       input-directory                        = 'data-files/incoming-files'
       processed-directory                    = 'data-files/processed-files/%Y-%m-%d/'
       rejected-directory                     = 'data-files/rejected-files/%Y-%m-%d/'
       access-model                           = '$(db-access-model)'
       database                               = 'glass-db'
       file-pattern                           = 'bloomberg_gain_TA.txt'
       delimiter                              = '|'
       header                                 = 'true'
       footer                                 = 'true'
       update-only-mode                       = 'true'
       date-format                            = 'dd-MMM-yy'
       allow-null-in-update-only-mode  = 'NEXT_DIVIDEND_ANNOUNCED_DATE,NEXT_DIVIDEND_CURRENCY_ID,NEXT_
DIVIDEND_EX_DATE,NEXT_DIVIDEND_AMOUNT,NEXT_DIVIDEND_PAY_DATE,NEXT_DIVIDEND_RECORD_DATE,NEXT_DIVIDEND_
ACTION_TYPE'
>
       <default-field name = 'SOURCE'          value = 'G1' />
       <default-field name = 'ISSUER_SOURCE'   value = 'G1' />

       <field-alias    name = 'SOURCE_ID'                      field = 'SECURITY_ID' >
       <field-alias    name = 'SHORT_NAME'                     field = 'SECURITY_DES' >
       <field-alias    name = 'ISSUED_AMOUNT'                  field = 'AMT_ISSUED' >
       <field-alias    name = 'OUTSTANDING_AMOUNT'             field = 'AMT_OUTSTANDING' >
       <field-alias    name = 'DIVIDEND_FREQUENCY'             field = 'DVD_FREQ' >
       <field-alias    name = 'GICS_SUB_INDUSTRY_ID'           field = 'GICS_SUB_INDUSTRY' >
       <field-alias    name = 'ISSUER_SOURCE_ID'               field = 'ISSUER' >
       <field-alias    name = 'MINIMUM_DENOMINATION'           field = 'MIN_DENOMINATION' >
       <field-alias    name = 'MINIMUM_INCREMENT'              field = 'MIN_INCREMENT' >
       <field-alias    name = 'PRIMARY_INDEX'                  field = 'REL_INDEX' >
       <field-alias    name = 'SHORT_INTEREST'                 field = 'SHORT_INT' >
       <field-alias    name = 'EXCHANGE_ID'                    field = 'EQY_PRIM_EXCH_SHRT' >
       <field-alias    name = 'RATING'                         field = 'RTG_SP' >
       <field-alias    name = 'NEXT_DIVIDEND_ANNOUNCED_DATE'   field = 'EQY_DVD_CASH_ANN_DT_NEXT' >
       <field-alias    name = 'NEXT_DIVIDEND_CURRENCY_ID'      field = 'EQY_DVD_CASH_CRNCY_NEXT' >
       <field-alias    name = 'NEXT_DIVIDEND_EX_DATE'          field = 'EQY_DVD_CASH_EX_DT_NEXT' >
       <field-alias    name = 'NEXT_DIVIDEND_AMOUNT'           field = 'EQY_DVD_CASH_GROSS_NEXT' >
       <field-alias    name = 'NEXT_DIVIDEND_PAY_DATE'         field = 'EQY_DVD_CASH_PAY_DT_NEXT' >
       <field-alias    name = 'NEXT_DIVIDEND_RECORD_DATE'      field = 'EQY_DVD_CASH_RECORD_DT_NEXT' >
       <field-alias    name = 'NEXT_DIVIDEND_ACTION_TYPE'      field = 'EQY_DVD_CASH_TYP_NEXT' >
</FINSecurityImport>
```

File Patterns and Directory Names

`file-pattern` is the name of the file which is imported if found in the `input-directory`. `file-pattern` can contain the special characters:

- * - match zero or more characters
- ? - match any single character

If the file is being delivered into the `input-directory` via FTP/SFTP, for example, then the file should be transferred with a temporary filename and then renamed to the filename matching `file-pattern` when the file transfer is complete. This is to avoid a situation in which the file is opened and read before the file transfer is complete.

Within loader definitions it is common to see the directory paths similar to:

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```
input-directory = 'data-files/incoming-files'
processed-directory = 'data-files/processed-files/%Y-%m-%d/'
rejected-directory = 'data-files/rejected-files/%Y-%m-%d/'
```

The `file-pattern` and directory paths can contain the same special data and time specifiers as the `log-filename` command line parameter. For a complete list of specifiers see Date and Time Format Specifiers.

CSV Without Header Row

Imported CSV files often contain field ids within the first row or record. However, it is possible to import CSV files which do not contain this header record.

Without a header record it's necessary to define the field id order in the loader definition.

This is achieved using the `include-column` and `field-list` elements. For example:

```
<GICSMap class = 'FINGICSMapDelimitedFile'
       ...
       include-column-definitions = 'false'
       field-list = 'GICS_CODE,DESCRIPTION'
/>
```

Documentation of the file formats supported can be found in the *Data Import Guide* Technical Manual.

Delimiters

By default the data import files are expected to have comma (',') field delimiters. In some cases other delimiter characters may be used. The delimiter character can be specified within the configuration of a file importer.

For example, here's how to specify a delimiter of a pipe character ('|') in a Global One Currency loader:

```
<Currency class = 'G1Currency'
       access-model = '$(db-access-model)'
       ...
       delimiter = '|'
       ...
```


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Database Connectivity

Oracle

The Oracle database name and user credentials are typically specified in the `database.xml` configuration file. For example:

```
<username>glass_svr</username>
<password>ta</password>
<database>$(GLASS_ORA_DB)</database>
```

The default password for database logins is 'xx'.

⚠ The default password should be changed immediately to something more secure.

Reconnection Settings

It is possible to define how the application will attempt to reconnect to the database when connectivity is lost. This is specified in the `database.xml` configuration file. For example:

```
<username>glass_svr</username>
<password>ta</password>
<database>$(GLASS_ORA_DB)</database>
<reconnect-attempts>5</reconnect-attempts>
<sleep-between-reconnect-attempts>10</sleep-between-reconnect-attempts>
<idle-timeout>5</idle-timeout>
<connection-pool-size>10</connection-pool-size>
```

Where:

reconnect-attempts is the maximum number of reconnection attempts to make, with the default being 60.

sleep-between-reconnect-attempts is the time in seconds to wait following a failed connection attempt before another attempt should be made; with the default being 10 seconds.

idle-timeout is the time in seconds to wait before disconnecting idle database connections; with the default being 60 seconds.

connection-pool-size is the maximum number of concurrent connections to the database; with the default being 25.

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Install Type

It is important to ensure the correct 'Install Type' is specified for your business. This setting is used to identify whether the installation is for a Broker Dealer or Agent Lender. The setting is specified in the `glass.xml` or `modules.xml` configuration file. For example:

```
<module name='app-core-$(module-access-model)'>
       <install-type>agent-lender</install-type>
       <database id='glass-db'>
               <include filename='../environments/$(GLASS_ENVIRONMENT)/database.xml' /
       </database>
<module>
```

`install-type` can be agent-lender or broker-dealer.

Workspace Process Mode

You can specify the browser behaviour when opening a new tab/window from an action, such as creating a trade from a source record. The setting is specified in the `glass.xml` configuration file. For example:

```
<workspace-process-mode value='{SINGLE|MULTI}' />
```

If the value is SINGLE, then opening a new tab/window from an action, will open it in the same process as the existing tab.

MULTI will open the tab/window in a new process. This is the default as it results in improved memory usage.

Exporter Destinations

On some browsers there is the option to set up a schedule to regularly email the selected data via the Exporter screen. The user can select from a list of predefined destinations, which are defined in the `glass.xml` configuration file and the `<core>` section. For example:

```
<output-folder path = 'data-files/outgoing-files' users = 'Admin' />
```

```
<output-folder path = '\\svr001\Files\Test\XYZ\' roles = 'System Admin' />
```

Now, when glass is restarted, and the Exporter opened from Grid Tools, when creating a new Destination, a 'Type' field will appear in the Destination Editor, with a dropdown offering 'File' or 'Email'. Once 'File' is selected, the user will be prompted for a Title and Destination text, and a 'File Path' dropdown will be available - this will reflect the relevant output-folder items from the `glass.xml` file. So, from the example above, a user 'Admin' will see `data-files/outgoing-files` as an option; anyone in the System Admin Role will see `\\svr001\Files\Test\XYZ\`.

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Exception Handling

Embedded Exception Handler

On Linux the `glassd` process will attempt to write a backtrace to the `glassd` log file if a process exception occurs.

To configure a Linux machine to produce a core dump alter the user limit by adding an entry to the file `/etc/security/limits.conf` such as:.

```
the_glassd_user - core unlimited
```

Note: edit the `the_glassd_user` as appropriate to the user that runs `glassd`. Updating this file will need root privilege.

Alternatively edit the `glassd` user `~/.bashrc` file and append the line:

```
ulimit -c unlimited
```

To check the current user limits use:

```
ulimit -a
```

Additionally, you will need to ensure that you have sufficient free disk space on the partition where `glassd` is installed, since the core file will be approx the size of the virtual memory size of `glassd`.

By default the embedded exception handler is enabled. To completely disable the embedded exception handler set the following environment variable in the configuration:

```
<enable-exception-dump value='false'/>
```

The message `Exception handler is enabled` will be displayed near the top of the `glassd` log file if the handler is enabled.

Snapshots

On Linux it is possible to capture a backtrace of every thread running in the `glassd` process on demand using a script called `companyx-debug-stacks.sh` (this script can be supplied on request). This may be useful to diagnose the cause of a hung process.

The script will create a file called `companyx-debug-stacks-gdb.log` in the current directory. This can be e-mailed to `companyx-service@companyx.com` for analysis.

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Database Creation

SQL Scripts

Within the primary release archive `sql-scripts` directory there should be the following sub-directories:

- `creation-scripts`. SQL scripts and Windows batch files or Unix shell scripts for each supported database product.
- `migration-scripts`. SQL scripts to migrate from one schema version to a subsequent version for each supported database product.
- `safety-scripts`. SQL scripts to make safe email server mailboxes, email addresses, tasks which may communicate outside and external system credentials. These may be useful if you wish to take a copy of a database and move it to another environment.
- `sanitize-scripts`. SQL scripts to anonymise Personally Identifiable Information (PII).

A customer specific release archive may contain a `sql-scripts` sub-directory:

- `client-migration-scripts`. Additional SQL migration scripts specific to a customer.

⚠ The supplied SQL scripts assume the database name is glass and therefore some scripts will need to be modified if the database name is different.

Database and GLASS Schema Creation

Oracle

Note: There is one `GLASS` schema per Oracle database, so each runtime environment (Development, UAT, Production) will require its own Oracle database.

To create the database and all the necessary database objects follow these steps.

- Create a new Oracle database using Oracle's Database Configuration Assistant.

  Select Template 'General Purpose or Transaction Processing'.

  Global Database Name: glass

  SID: glass

- Open a Windows command prompt.
- Change directory to `sql-scripts\creation-scripts\oracle` (within the uncompressed release archive).
- Before the `GLASS` schema is created, it is necessary to create the required tablespaces.

  Edit `core\create_tablespaces.sql` and complete the `TODO` datafile locations. The location of the tablespaces will depend upon your specific environment and database server operating system.

  For example on Windows:

  ```
  CREATE TABLESPACE GLASS_DATA DATAFILE '{{TODO: path-to-database-datafiles}}/GLASS_DATA_01.DBF'
  ```

  might become:

  ```
  CREATE TABLESPACE GLASS_DATA DATAFILE 'C:\TA\GLASS_DATA_01.DBF'
  ```


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On Unix the `DATAFILE` path might be something like:

```
/u01/oracle/ica/glassdata.dbf
```

Execute the `create_tablespaces.sql` script to create the new tablespaces.

Oracle's `sqlplus` command line utility can be used to execute SQL scripts.

To run a script the syntax is:

```
sqlplus userid/password@database [AS SYSDBA] @script.sql
```

So, to run the `create_tablespaces.sql` script the command would be similar to:

```
sqlplus SYS/password@glass AS SYSDBA @create_tablespaces.sql
```

`SYS` is the name of the standard Oracle DBA account. Replace `password` will the correct password for user `SYS`.

- Set the environment variable `GLASS_ORA_DB` to the name of the newly created Oracle database.
- Edit the `create_schema` script and amend the username, password, and database name used by the **sqlplus** commands (this script needs to be executed by a user with the privilege to create roles and users).
- Edit the `create_tables` script and amend the username, password, and database name used by the **sqlplus** commands should they be required to be different from the default.
- Run the `create-schema` script.

This script will create the required roles and users, tables, sequences, constraints and indexes.

If the `create-schema` script is successful a schema called `GLASS` containing tables such as `GLASS.FIN_ACCOUNT` and `GLASS.FIN_CURRENCY` will have been created within the Oracle database.

Execution log files, which may include any error messages, will be written to `sql-scripts\creation-scripts\obdc\core` and also alongside the SQL scripts in `odbc` sub-directories such as `sql-scripts\creation-scripts\oracle\fin`.

Purpose of TA Users

The primary purpose of each of the TA Users is as follows:

- GLASS is for running SQL scripts to create the database schema, and for running migration scripts.
- GLASS_SVR is for the glassd process to connect to the GLASS database and read and update the table data.
- GLASS_READ is for the customer to query the database (if needed) from a non-application process, in read-only fashion.

The important distinction between GLASS and GLASS_SVR, is that only the GLASS user can perform DDL such as creating and dropping tables. In the `sql-scripts/creation-scripts/odbc/create_roles_and_users.sql` this can be seen in the lines:

```
GRANT CONNECT TO GLASS

GO

GRANT CONTROL TO GLASS

GO
```

The `glassd` process should not be able to create/drop tables, and so for enhanced security, does not have such permissions.

GLASS Schema Recreation

Oracle

The scripts `drop-schema`, `create-schema` and `recreate-schema` can be used to recreate the database schema without dropping and recreating the database. This allows users without the privilege to create a database the ability to recreate the database schema (permission to create databases is typically restricted to a limited number of accounts).

Restoration from Backup

Oracle

To restore a database from a backup use a tool such as `imp` or Oracle's Recovery Manager (RMAN).

Recreate the default database users and roles by running `sql-scripts\creation-scripts\oracle\core\create_roles_and_users.sql` before running `imp`.

For example:

```
imp 'SYS/password@glass AS SYSDBA' file=glass.dmp full=y
```

Important: If the backup was created whilst the database was in use, then the backup may be inconsistent with regard to the Oracle sequences and the table data. To rectify this run the sql script sql-scripts/creation-scripts/oracle/sequence-reset.sh

Using a Production Database in QA

A production database contains email addresses, and data-feed credentials. If a copy of a production database is used within QA, there is the potential for emails to be sent to these addresses, and cause confusion. To prevent this, a SQL script exists to put the database in a "safe" state. This script is sql-scripts/safety-scripts/oracle/make-safe.sql.

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Database Migration

SQL Scripts

Within the primary release archive `sql-scripts` directory there should be the following sub-directories:

- `creation-scripts`. SQL scripts and Windows batch files or Unix shell scripts for each supported database product.
- `migration-scripts`. SQL scripts to migrate from one schema version to a subsequent version for each supported database product.
- `safety-scripts`. SQL scripts to make safe email server mailboxes, email addresses, tasks which may communicate outside and external system credentials. These may be useful if you wish to take a copy of a database and move it to another environment.
- `sanitize-scripts`. SQL scripts to anonymise Personally Identifiable Information (PII).

A customer specific release archive may contain a `sql-scripts` sub-directory:

- `client-migration-scripts`. Additional SQL migration scripts specific to a customer.

⚠ The supplied SQL scripts assume the database name is glass and therefore some scripts will need to be modified if the database name is different.

Determine Current Schema Version

To determine the current version of the application database, if unknown, run the SQL:

```
select * from GLASS.SCHEMA_VERSION
```

The result will contain the current schema version. For example:

MAJOR	MINOR	REVISION	HOTFIX
3	0	0	0

The schema version doesn't change when a new revision or hotfix are released so `MAJOR` and `MINOR` are currently the only significant columns.

Schema Upgrade

The system upgrade is designed to operate on a single schema version upgrade. If multiple schema version upgrades are required, then it is necessary to complete all the following instructions at each individual version upgrade. Each step in the upgrade should complete successfully before continuing with the next step.

⚠ Before upgrading a database schema a backup should always be made.

Migration scripts and batch files should be run in the following order:

- First run any scripts in the `migration-scripts` directory e.g. `v3.x-to-v3.y.sql`.
- Check for a `client-migration-scripts` directory which will contain any additional scripts which should be run. These scripts will have matching version numbers for this schema version upgrade e.g. v3.x-to-v3.y.sql or v3.x-to-v3.y.bat.

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- Check for a `client-examples/banka` directory for any additional upgrade loaders that are applicable e.g. the directory `client-examples/banka/3.x-3.y-migration/`. If the relevant directory is present then it will have the appropriate install-oracle.sh script.

If each of the above steps completed successfully, then the system should now be ready to use.

Note that if the upgrade is within a QA environment you may want to run the "safety-scripts". See Using a Production Database in QA.

Oracle

To upgrade the database schema follow these steps for every batch file or script in each of the `migration-scripts`, `client-migration-scripts` and `client-examples` directories.

- Open a Windows command prompt.
- Change to the appropriate directory. For example: `sql-scripts\migration-scripts\oracle` (within the uncompressed release archive).
- Run the command:

```
sqlplus GLASS/password@glass @v3.x-to-v3.y.sql
```

This command assumes that the database is called `glass`. The correct SYS password will need to be specified in place of `password`.

Substitute the actual SQL migration script for `v3.x-to-v3.y.sql`. For example, to migrate from version 3.7 to 3.8 the SQL migration script would be called `v3.7-to-v3.8.sql`.

If the script is successful and runs to completion the row in table `SCHEMA_VERSION` will be changed to the new version. Re-run the SQL: `select * from GLASS.SCHEMA_VERSION`to check that the version has been updated.

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Date and Time Format Specifiers

A list of all the date and time format specifiers which can be used in the `log-filename` command line argument and loader directory names. They all begin with a percentage ('%') sign.

Specifier	Replaced By	Example
%A	Full weekday name	Thursday
%B	Full month name	August
%d	Day of the month, zero-padded (01-31)	12
%e	Day of the month, space-padded (1-31)	28
%F	Short YYYY-MM-DD date, equivalent to %Y-%m-%d	2001-08-23
%H	Hour in 24h format (00-23)	14
%I	Hour in 12h format (01-12)	02
%j	Day of year as decimal number (001-366)	235
%m	Month as a decimal number (01-12)	05
%M	Minute (00-59)	55
%p	AM or PM	PM
%S	Second (00-61)	02
%u	ISO 8601 weekday as number with Monday as 1 (1-7)	4
%V	ISO 8601 week number (00-53)	34
%w	Weekday as a decimal number with Sunday as 0 (0-6)	3
%y	Year, last two digits (00-99)	14
%Y	Year	2014

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Software Archive Download

Files can be transferred to and from the `ftp.companyx.com` file server using a variety of protocols. These include HTTP, FTPS, SFTP, FTP and WebDav.

Please contact *Support* if you have difficulty using your file download account.

HTTPS

Host: ftp.companyx.com

Port: 443

All major web browsers including Internet Explorer, Firefox and Chrome are supported.

FTPS

Host: ftp.companyx.com

Port: 21 or 10021

Note that your file transfer client should use the option `FTP with Explicit SSL/TLS`.

SFTP

Host: ftp.companyx.com

Port: 22 or 12222

The host key should match

FTP

Host: ftp.companyx.com

Port: 10221

Note that your file transfer client should use the option `FTP with Implicit SSL/TLS`.

If possible use the alternative service of `FTPS with Explicit SSL/TLS` on port 21 or 10021.

WebDav

Host: ftp.companyx.com

Port: 443

To connect using Windows File Explorer map a network drive specifying the Folder as:

`https://ftp.companyx.com`

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Automatic Archiving

The application supports automatic data purging once records have reached a certain age in several tables. These include:

- Benchmarks such as Astec, Markit and Internal
- External Positions
- External Trades
- Indicative Rates
- Indicative Utilization
- Push Lists
- Shortlist Lists
- Collateral Sweep Lists

The number of days is configurable on a per table basis. Audit records are not automatically purged. You may also regularly run your own SQL scripts as part of general housekeeping.

Automatic Archiving is switched on using the `glass.xml` environment setting:

```
{code}
      <enable-auto-archiving value='true' />
{code}
```

To override the purge days the environment settings are:

Table	Setting
Astec	`<astec-seclen-data-archive-data-after-days>`
Markit	`<markit-secfin-data-archive-data-after-days>`
External Positions	`<inv-external-position-archive-data-after-days>`
External Trade	`<inv-external-trade-archive-data-after-days>`
Push Lists lists	`<pushlist-list-archive-data-after-days>`
Specialness Event	`<inv-specialness-event-archive-data-after-days>`
Indicative Rates	`<fin-indicative-rate-archive-data-after-days>`
Daily Aggregate	`<eqlend-daily-aggregate-archive-data-after-days>`
Internal Positions	`<inv-internal-position-archive-data-after-days>`
Internal Security Financial Data	`<fin-internal-sec-fin-data-archive-data-after-days>`
Benchmark Rates	`<fin-benchmark-rate-archive-data-after-days>`
Indicative Utilisation	`<fin-indicative-utilization-archive-data-after-days>`
Sovereign Benchmark Rates	`<sov-bm-rate-archive-data-after-days>`

A purge days of an empty string or alternatively no purge days configuration will result in a coded default purge days setting of 2 * 365 days being applied (subject to auto-archiving being enabled).

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SSL Certificate Generation

The SSL certificate can be either a self-signed certificate or a Certificate Authority certificate, however if a self-signed certificate is used then users may be warned that there are potential security risks unless the self-signed certificate is installed as a trusted certificate on every client machine that accesses the application. Consequently, whilst self-signed certificates are ideal for test environments they are not recommended for production.

To Generate a Self-Signed Certificate

Generating a self-signed certificate is quick, simple and free of charge.

Depending on your organisational policy you may need to obtain your IT department's root certificate or even generate your own, but for simplicity we won't use one in our example. To generate a self-signed certificate using OpenSSL:

- Download and install OpenSSL from https://www.openssl.org, if you don't have it already installed.
- Run the following from the shell or command prompt:

```
openssl req -x509 -days 365 -nodes -newkey rsa:2048 -out hostname.crt -outform pem -keyout
hostname.key -keyform pem
```

- Copy both of the files which are created into the `configuration/environments/common/` directory located within your installation.
- Modify the environment configuration file to ensure that the http and ssl configuration sections are included and the certificate and private-key values are set correctly. For details of how to modify the configuration file see HTTP Server.

To Generate a Certificate using a Certificate Authority (CA)

It is recommended to use a certificate from a trusted CA such as Verisign or GlobalSign.

The process can take a number of weeks depending on the certificate type you choose, the level of warranty / insurance you require, the level of validation which has to be performed and which CA you choose. Domain validated certificates with no or limited warranty can take less time compared to those that included extended validation with larger warranties.

In addition to the level of validation and warranty, you should also consider what type of certificate you require. For example:

A **Single Server Certificate** is only valid for the server name that it is installed on. For example: server.example.com. If you relocate the application to another server you will need to obtain a new certificate. Additionally if you have an alias to the server, for example the application is installed on server001.example.com but your users will be using https://ta.example.com to access the application, then you will need to use the alias hostname as the fully qualified domain name when generating the certificate request or use a SAN or Wildcard certificate which allow additional hostnames to be specified.

A **SAN Certificate** allows you define several alternative names in addition to the main hostname for that server. These could be for (but are not limited to) the same domain. For example: server001.example.com, server002.example.com, ta.example.com and ta.example.net. Depending on the type of SAN certificate you purchase you could include between 1 and 4 alternatives, 5 and 9 alternatives or even more. Typically the more alternatives you add the higher the cost of these certificates.

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A **Wildcard Certificate** allows you to install the certificate on any machine within the domain for which it is valid. For example a *.example.com certificate would be valid for server001.example.com, server002.example.com or ta.example.com but not ta.example.net

If you decide to use either a SAN or wildcard certificate and install it on multiple servers, then you will need to re-install the new certificate on all servers if:

- The certificate expires
- The certificate is revoked by the CA or any of the intermediaries are revoked
- The security of the certificate, or any of the intermediaries, are compromised

To generate a request for a CA certificate (CSR) using OpenSSL:

- Download and install OpenSSL if you don't have it already installed
- Run the following from the shell or command prompt:

```
openssl req -new -newkey rsa:2048 -nodes -out hostname.csr -outform pem -keyout hostname.key -
keyform pem
```

- Copy the private key file which is created into the `configuration/environments/common/` directory located within your installation.
- Send the CSR to the Certificate Authority
- Once the certificate request has been approved and the certificate generated, download the certificate from the CA
- Copy the certificate file to the `configuration/environments/common/` directory located within your installation.
- If intermediary certificates are included in the download then these will need to be appended to the certificate file that is used.
- You may also need to include the CA's root certificate in the certificate that is used.
- Modify the environment configuration file to ensure that the `http` and `ssl` configuration elements are included and the certificate and private key values are set correctly. For details of how to modify the configuration file see HTTP Server.

Notes about the command examples

hostname.csr, hostname.crt, hostname.key are the request, certificate and key filenames, and we would recommend using the hostname of your server rather than "hostname" in the filename. For example: server001.example.com or localhost.

The –nodes option generates a private key without a passphrase. If you would like to include a passphrase please leave this option out and add the `httpd/ssl/passphrase` configuration option to the environment configuration file.

The rsa:2048 options tells OpenSSL that you want to use 2048-bit RSA encryption. Depending on your IT policy you may want to change this to 2048 or 4096.

In our example we have set the validity period of the certificate to one year (365 days). If you would prefer to use a longer or shorter period then you can adjust this option accordingly.

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You may also want to sign the certificate with a specific digest. To do this you can use the `-md5, -sha1, -md2, -mdc2 and -md4` parameters.

Depending on your installation you may need to specify where the OpenSSL config file is located, if you received an error such as `"Unable to load config info from C:\OpenSSL/ssl/openssl.cnf"`. To do this either copy the config file from the OpenSSL apps directory to the `ssl` directory (if you have one), copy it to your HOME directory (under Linux) or add the following option when you run OpenSSL from the shell or command prompt. For example: `-config \openssl\apps\openssl.cnf`

Required Information

During the certificate request process you will be required to input several pieces of information. This information is normal to the certificate request process and will be embedded in the final certificate that is generated. Users will be able to see this information if they examine the certificate from their browser so we would recommend using accurate information.

Country Name

This is the two letter ISO code for your organisation's location. If your organisation is multi-national then you could use the location of the main office, head quarters or IT department. Other locations could include your office or your department's office - for example: US or UK.

State or Province Name

This is the state or province for your organisation's or department's location - for example: New York or Greater London.

Locality Name

This is the city where your organisation or department is located - for example: New York or London.

Organization Name

This is the legal name of your organisation and should be formatted according to your organisation's policy - for example: Your Company PLC.

Organizational Unit Name

This is the organisational unit or department name - for example: Head Office or IT.

Common Name

This is the fully qualified name of your server - for example: server001.example.com or localhost.

Email Address

This is the contact email address of the department or person that is responsible for SSL certificates within your organisation - for example: `it@example.com` or `fred.bloggs@example.com`.

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Troubleshooting

This section helps you to identify and resolve basic problems you may be experiencing with installing the product.

If you need assistance with any of the steps, or you cannot resolve your issue, please contact Support.

General software issues

UI is Unresponsive

There can be a number of issues that might result in this, but essentially it is not possible to access the http interface, despite the server continuing to process inbound feeds successfully. This may be caused by a lock being held on threads that are accessing the http interface; a lock which may only be released after a considerable period of time. When this occurs, refer to Support, providing the log files.

If this becomes a regular occurrence, then add to the configuration file, in the `<environment>` section, `<lock-enable-held-timing>true</lock-enable-held-timing>`. Once restarted, this will provide extra debugging information. Note that this should not be enabled on a permanent basis due to the volume of logging that is produced.

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European Sales Office

xxxxxxxxxxxxx

Tel: xxxxxxxx

North American Sales Office

xxxxxxxxxxx,

xxxxxxxxxxxx

Tel: xxxxxxxxxxxxx

Email: sales@companyx.com